THE LOCALISATION S^{-1}A IS THE ZERO RING EXACTLY WHEN 0 LIES IN S

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The equivalence relation defining the localisation. Definition of S^{-1}A, used at proof line 1333 to identify a pair with (0, 1). Gives (a, s) equivalent to (0, 1) whenever 0 lies in S, which makes every class equal.
- A ring with $1 = 0$ is the zero ring. Elementary, used at proof line 1334 in the contrapositive direction. Turns $1/1$ different from $0/1$ into the statement that S^{-1}A is not the zero ring.

Lemma 0.1. *The localization $S^{-1}A$ is the zero ring if and only if $0 \in S$.*

Proof. If $0 \in S$, any pair $(a, s) \sim (0, 1)$ by definition. If $0 \notin S$, then clearly $1/1 \neq 0/1$ in $S^{-1}A$. □

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